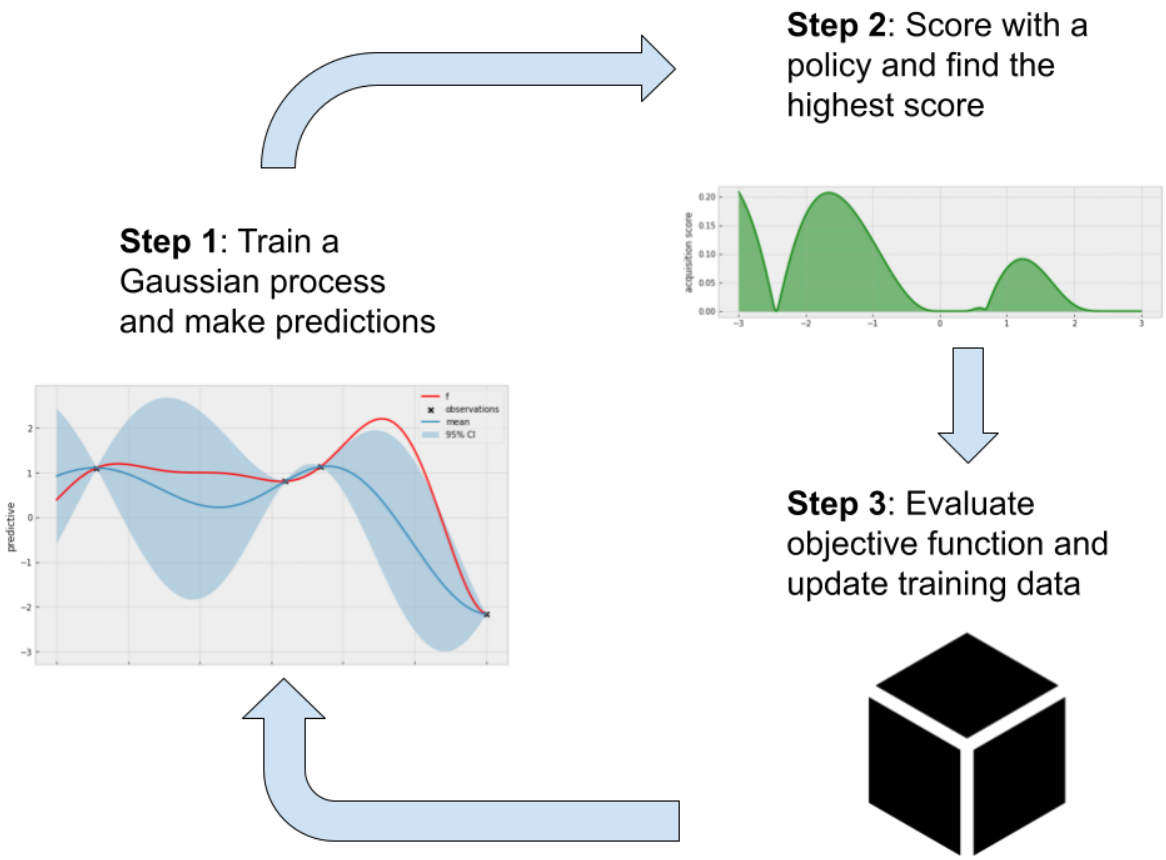


Figure 4.2 The Bayesian optimization loop, which combines a GP for modeling and a policy for decision making. This complete workflow may now be used to optimize black-box functions.

We repeat this loop until we reach a termination condition, most often once we have evaluated the objective function for a targeted number of loop iterations. This procedure is end-to-end, as we don't just use a GP to make predictions for the sake of making predictions—these predictions inform the decisions we make about which data points to collect next, which in turn drive how future predictions are generated.

NOTE	**The Bayesian optimization loop is a virtuous cycle of model training (the Gaussian process) and data collection (the policy), each of which helps and benefits from by the other, with the goal of locating the optimum of the objective function.**

SIDEBAR	**SIDEBAR** **The virtuous cycle of BayesOpt is to be contrasted with a vicious cycle, whose components aggravate one another to result in a bad outcome.**

The rule that dictates how data points are scored according to their value in helping us achieve this goal is decided by a BayesOpt *policy*, and is therefore essential to optimization performance.

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